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Inventor(s)

CLARK; Daniel et al.

CONTAINER-HANDLING SYSTEM

Abstract

A load handling device is provided for lifting and moving storage containers arranged in stacks in a storage structure, the storage structure comprising a track structure above the stacks of storage containers. The load handling device comprises a body, a driving assembly configured to move the body on the track structure, and a container-handling system. The container-handling system comprises: a holding frame; a holding assembly mounted on the holding frame, wherein the holding assembly is configured to releasably hold a storage container in a holding region with respect to the holding frame; a lifting assembly configured to lower and raise the holding frame relative to the body; and an overheight detection system comprising object detection means mounted on the holding frame, wherein the object detection means is configured to detect the presence of an object in an overheight region, wherein the overheight region is above the holding region.

Inventors: CLARK; Daniel (Hatfield, GB), RUFFELL; Alan (Hatfield, GB), GRUBER; Malte (Stockholm, SE)

Applicant: Ocado Innovation Limited (Hatfield, GB)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This Application is a continuation of PCT International Patent Application No. PCT/EP2023/083154, filed on Nov. 27, 2023, which claims priority to UK Patent Application No. GB2217791.9, filed on Nov. 28, 2022, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The present invention relates to devices for handling storage containers in a storage and retrieval system.

BACKGROUND

[0003] Some commercial and industrial activities require systems that enable the storage and retrieval of a large number of different products. WO2015019055A1 describes a storage and retrieval system in which items are stored in storage containers and the storage containers are arranged in stacks within a grid storage structure. The system further comprises remotely operated load handling devices configured to move on tracks located on the top of the grid storage structure. To pick up or drop off storage containers stored in the grid storage structure, each load handling device is equipped with a gripper device for releasably holding a storage container, and a lifting assembly for raising and lowering the gripper device.

[0004] Objects that have not been loaded in an expected manner into the storage containers can cause problems within the storage and retrieval system. For example, a situation may occur in which an object protrudes above the top of a storage container. A protruding item can cause a number of issues, such as preventing the gripper device from engaging properly with the storage container and preventing the storage container from being properly received by the load handling device, as well as downstream issues such as the storage container becoming stuck in another region of the storage and retrieval system.

[0005] The present invention aims to address the problem of storage containers containing protruding items within a storage and retrieval system.

SUMMARY

[0006] The invention is defined in the accompanying claims.

[0007] The invention provides a container-handling system comprising: a holding frame; a holding assembly mounted on the holding frame, wherein the holding assembly is configured to releasably hold a storage container in a holding region with respect to the holding frame; and an overheight detection system comprising object detection means mounted on the holding frame, wherein the object detection means is configured to detect the presence of an object in an overheight region, wherein the overheight region is above the holding region.

[0008] The overheight region may be substantially planar. The overheight region may be a substantially horizontal plane. This helps to minimize the chance of erroneously detecting an object within the holding region (i.e. within the storage container). The object detection means may be

configured to detect the presence of an object between at least two points lying in the horizontal plane. The overheight region may be at a fixed position with respect to the holding frame. The object detection means may be configured to detect an object extending from the holding region into or through the overheight region.

[0009] The holding region may be at a fixed position with respect to the holding frame. The holding assembly may be configured to releasably hold a storage container from above. The holding region may be below the holding frame. The holding frame and the holding assembly may be configured so as to define an upper limit of the holding region. In other words, the holding frame and the holding assembly may be configured such that a storage container being held by the holding assembly cannot move upwardly relative to the holding frame past the upper limit due to being physically blocked by a component of the holding frame. The holding frame may comprise one or more frame members arranged in a substantially horizontal plane. The holding frame may have a rectangular shape. The holding region may be below the frame members. An upper limit for the holding region may be the bottom-most surface of the frame members. The overheight region may be above the bottom-most surface of the frame members.

[0010] The holding frame may comprise a frame opening extending vertically through the holding frame. At least a portion of the overheight region may be located within the frame opening. The overheight region may be fully within the frame opening. At least a portion of the overheight region may be located above the frame opening. The overheight region may be fully above the frame opening. At least a portion of the object detection means may be mounted within the holding frame opening. At least a portion of the object detection means may be mounted above the frame opening. The object detection means may be configured to detect an object extending from the holding region into or through the frame opening.

[0011] The object detection means may comprise a sensor assembly arranged on opposing lateral sides of the overheight region. In the case where the holding frame comprising a frame opening, the sensor assembly may be arranged on opposing lateral sides of the frame opening. The sensor assembly may comprise one or more photoelectric sensors. The sensor assembly may comprise a through-beam sensor arrangement. The sensor assembly may comprise a retroreflective sensor arrangement.

[0012] The sensor assembly may comprise at least one transmitter and at least one receiver mounted on the holding frame on opposing lateral sides of the overheight region. The at least one transmitter may be configured to emit a signal towards the at least one receiver. The at least one receiver may be configured to detect the signal from the at least one transmitter. The overheight detection system may further comprise a controller communicably coupled to the at least one transmitter and the at least one receiver. The controller may be configured to determine that an object is in the overheight region when the signal between the at least one transmitter and the at least one receiver is interrupted.

[0013] The sensor assembly may comprise at least one transmitter and at least one receiver mounted on the holding frame on the same lateral side of the overheight region, and at least one reflector at an opposing lateral side of the overheight region. The at least one transmitter is configured to emit a signal towards the at least one reflector and the at least one receiver is configured to detect the signal from the at least one transmitter after being reflected by the at least one reflector. The overheight detection system may further comprise a controller communicably coupled to the at least one transmitter and the at least one receiver. The controller may be configured to determine that an object is in the overheight region when the signal between the at least one transmitter and the at least one receiver via the at least one reflector is interrupted.

[0014] The object detection means may comprise a camera mounted on the holding frame. The camera may be configured to capture an image containing the overheight region. The object detection means may further comprise one or more processors configured to analyze the image and determine the presence of an object within the overheight region.

[0015] The object detection means may comprise a substantially horizontal sheet defining the overheight region. The object detection means may be configured to detect a force being applied to the horizontal sheet to determine the presence of an object in the overheight region. The object detection means may comprise a mechanical force sensor.

[0016] The container-handling system may further comprise a lifting assembly configured to lower and raise the holding frame.

[0017] The lifting assembly may comprise at least one tether and at least one motor or other winding and unwinding means. The at least one tether may be (directly or indirectly) coupled between the holding frame and the at least one motor. The at least one motor may be configured to wind and unwind the at least one tether to raise and lower the holding frame respectively.

[0018] The container-handling system may further comprise a control system comprising one or more controllers configured to control operation of the holding assembly based on a detection outcome by the object detection means. The control system may be configured such that the holding assembly is not activated if the object detection means detects the presence of an object in the overheight region.

[0019] In the case where the container-handling system comprises the lifting assembly, the control system may be configured to activate the overheight detection system (i.e. activate the object detection means) when the lifting assembly is lowering the holding frame, or once the lifting assembly has lowered the holding frame to a predetermined position.

[0020] In the case where the container-handling system comprises the lifting assembly, the control system may be configured to control the lifting assembly to raise the holding frame without activating the holding assembly if the object detection means detects the presence of an object in the overheight region.

[0021] The overheight detection system may be configured to determine the height of an object protruding above a storage container based on a vertical distance descended by the holding frame during a time period between the point at which an object is first detected in the overheight region and the point at which a storage container is in the holding region.

[0022] The holding assembly may comprise a plurality of grippers mounted on the holding frame. Each gripper may be movable between an engaged position and a disengaged position for engaging and disengaging the storage container respectively.

[0023] The present invention also provides a load handling device for lifting and moving storage containers arranged in stacks in a storage structure, the storage structure comprising a track structure above the stacks of storage containers. The load handling device may comprise: a body; a driving assembly configured to move the body on the track structure; and the container-handling system having a lifting assembly as defined above, wherein the lifting assembly is configured to lower and raise the holding frame relative to the body.

[0024] The load handling device may further comprise a container-receiving space configured to accommodate a storage container held by the holding frame. The lifting assembly may be configured to raise and lower the holding frame into and out of the container-receiving space respectively.

[0025] The track structure may be a track structure comprises a first set of tracks and a second set of tracks, the first set of tracks extending in a first direction and the second set of tracks extending in a second direction, the second direction being substantially perpendicular to the first direction, to form a grid pattern defining a plurality of grid cells. The driving assembly may be configured to move the body of the load handling device on this track structure.

[0026] The present invention also provides a storage and retrieval system comprising: a storage structure comprising: a plurality of stacks of storage containers, each storage container having a top opening through which objects can be placed into the storage container; and a track structure arranged above the stacks of storage containers and configured to allow access to each stack from above, the storage and retrieval system further comprising at least one load handling device as

defined above, wherein the container-handling system is configured to pick up and drop off storage containers from and onto the stacks of storage containers.

[0027] The track structure may comprise a first set of tracks and a second set of tracks. The first set of tracks may extend in a first direction and the second set of tracks extending in a second direction, the second direction being substantially perpendicular to the first direction, to form a grid pattern defining a plurality of grid cells. Each stack of storage containers may be arranged below a grid cell.

[0028] The storage and retrieval according may further comprise a central control system configured to communicate with the at least one load handling device. The central control system may be configured to command the load handling device to move to a target grid cell from a list of available grid cells to pick up or drop off a storage container below the target grid cell. The central control system may be further configured to remove the target grid cell from the list of available grid cells if the overheight detection system detects an object in the overheight region when the load handling device is at the target grid cell.

[0029] The present invention also provides a method of using the container-handling system defined above. The method comprises the steps of: lowering the holding frame towards a storage container; detecting whether or not an object is present within the overheight region; if an object is detected, raising the holding frame away from the storage container without activating the holding assembly; and if an object is not detected, activating the holding assembly to hold the storage container and then raising the holding frame.

Description

DESCRIPTION OF THE DRAWINGS

[0030] The present invention will now be described, by way of example only, with reference to the accompanying drawings, in which like reference numerals are used for like features, and in which:

[0031] FIG. 1 is a schematic perspective view of a grid storage structure and containers arranged within the grid storage structure.

[0032] FIG. 2 is a schematic plan view of a track structure on top of the storage structure of FIG. 1.

[0033] FIG. 3 shows load handling devices on top of the track structure of the storage structure of FIG. 1.

[0034] FIG. 4 is a schematic perspective view of a load handling device with a holding frame in a position below the bottom of the load handling device.

[0035] FIG. 5 is a schematic perspective view of the load handling device of FIG. 4 with a side panel removed to show a container-receiving space.

[0036] FIG. 6 is a schematic perspective view of the load handling device of FIG. 5 with a container occupying the container-receiving space.

[0037] FIG. 7 is a perspective view of a container-handling system comprising a lifting assembly and a holding frame.

[0038] FIG. 8A is a front perspective view of a gripper assembly in a closed configuration. FIG. 8B is a front perspective view of a gripper assembly in an open configuration.

[0039] FIG. 9A is a rear perspective view of the gripper assembly in a closed configuration, as in FIG. 8A. FIG. 9B is a rear perspective view of the gripper assembly in an open configuration, as in FIG. 8B.

[0040] FIG. 10 is a perspective view of a storage container that may be held by the holding frame of the container-handling system.

[0041] FIG. 11A is a side view of the holding frame and the storage container with the gripper assembly disengaged with the storage container. FIG. 11B is a side view of the holding frame and the storage container with the gripper assembly engaged with the storage container.

[0042] FIG. **12** is a perspective view of the holding frame holding the storage container.

[0043] FIG. **13** is a schematic perspective view of the holding frame holding the storage container and an example object detection means of the container-handling system.

[0044] FIG. **14** is a schematic cross-sectional view of the example of FIG. **13** with an overheight object in the storage container.

[0045] FIG. **15** is a schematic perspective view of the holding frame holding the storage container and another example object detection means of the container-handling system.

[0046] FIG. **16** is a schematic perspective view of the holding frame holding the storage container and another example object detection means of the container-handling system.

[0047] FIG. **17** is a schematic bottom perspective view of an alternative holding frame of the container-handling system.

[0048] FIG. **18** is a block diagram of a control system.

DETAILED DESCRIPTION

[0049] FIG. **1** illustrates an example storage structure **1** that may be used in a storage and retrieval system to store storage containers **50**. The storage structure **1** comprises a holding framework comprising upright members **3** and horizontal members **5**, **7** which are supported by the upright members **3**. The horizontal members **5** extend parallel to one another and the illustrated x-axis. The horizontal members **7** extend parallel to one another and the illustrated y-axis, and transversely to the horizontal members **5**. The upright members **3** extend parallel to one another and the illustrated z-axis, and transversely to the horizontal members **5**, **7**. The horizontal members **5**, **7** form a grid pattern defining a plurality of grid cells **14**. In the illustrated example, storage containers **50** are arranged in stacks **11** beneath the grid cells **14** defined by the grid pattern, one stack **11** of storage containers **50** per grid cell **14**.

[0050] FIG. **2** shows a large-scale plan view of a section of track structure **13** forming part of the storage structure **1** illustrated in FIG. **21** and located on top of the horizontal members **5**, **7** of the storage structure **1** illustrated in FIG. **1**. The track structure **13** may be provided by the horizontal members **5**, **7** themselves (e.g. formed in or on the surfaces of the horizontal members **5**, **7**) or by one or more additional components mounted on top of the horizontal members **5**, **7**. The illustrated track structure **13** comprises x-direction tracks **17** and y-direction tracks **19**, i.e. a first set of tracks **17** which extend in the x-direction and a second set of tracks **19** which extend in the y-direction, transverse to the tracks **17** in the first set of tracks **17**. The tracks **17**, **19** define apertures **15** at the centers of the grid cells **14**. The apertures **15** are sized to allow storage containers **50** located beneath the grid cells **14** to be lifted and lowered through the apertures **15**. The x-direction tracks **17** are provided in pairs separated by channels **21**, and the y-direction tracks **19** are provided in pairs separated by channels **23**. Other arrangements of track structure may also be possible.

[0051] FIG. **3** shows a plurality of load handling devices **25** moving on top of the storage structure **1** illustrated in FIG. **1**. The load handling devices **25**, hereinafter referred to as “bots”, are provided with sets of wheels to engage with corresponding x-or y-direction tracks **17**, **19** to enable the bots **25** to travel across the track structure **13** and reach specific grid cells **14**. The illustrated pairs of tracks **17**, **19** separated by channels **21**, **23** allow bots **25** to occupy (or pass one another on) neighboring grid cells **14** without colliding with one another.

[0052] As illustrated in FIG. **4**, a bot **25** comprises a body **27** in or on which are mounted one or more components which enable the bot **25** to perform its intended functions. These functions may include moving across the storage structure **1** on the track structure **13** and raising or lowering storage containers **50** (e.g. from or to stacks **11**) so that the bot **25** can retrieve or deposit storage containers **50** in specific locations defined by the grid pattern.

[0053] The illustrated bot **25** comprises a driving assembly comprising first and second sets of wheels **29**, **31** which are mounted on the body **27** of the bot **25** and enable the bot **25** to move in the x-and y-directions along the tracks **17** and **19**, respectively. In particular, two wheels **29** are provided on the shorter side of the bot **25** visible in FIG. **4**, and a further two wheels **29** are

provided on the opposite shorter side of the bot 25. The wheels 29 engage with tracks 17 and are rotatably mounted on the body 27 of the bot 25 to allow the bot 25 to move along the tracks 17. Analogously, two wheels 31 are provided on the longer side of the bot 25 visible in FIG. 4, and a further two wheels 31 are provided on the opposite longer side of the bot 25. The wheels 31 engage with tracks 19 and are rotatably mounted on the body 27 of the bot 25 to allow the bot 25 to move along the tracks 19.

[0054] To enable the bot 25 to move on the different wheels 29, 31 in the first and second directions, the driving assembly further comprises a wheel-positioning mechanism (not shown) for selectively engaging either the first set of wheels 29 with the first set of tracks 17 or the second set of wheels 31 with the second set of tracks 19. The wheel-positioning mechanism is configured to raise and lower the first set of wheels 29 and/or the second set of wheels 31 relative to the body 27, thereby enabling the load handling device 25 to selectively move in either the first direction or the second direction across the tracks 17, 19 of the storage structure 1.

[0055] The wheel-positioning mechanism may include one or more linear actuators, rotary components or other means for raising and lowering at least one set of wheels 29, 31 relative to the body 27 of the bot 25 to bring the at least one set of wheels 29, 31 out of and into contact with the tracks 17, 19. In some examples, only one set of wheels is configured to be raised and lowered, and the act of lowering the one set of wheels may effectively lift the other set of wheels clear of the corresponding tracks while the act of raising the one set of wheels may effectively lower the other set of wheels into contact with the corresponding tracks. In other examples, both sets of wheels may be raised and lowered, advantageously meaning that the body 27 of the bot 25 stays substantially at the same height and therefore the weight of the body 27 and the components mounted thereon does not need to be lifted and lowered by the wheel-positioning mechanism.

[0056] FIG. 4 also schematically shows a container-handling system 100 for holding and transporting storage containers 50 between different locations in the storage structure 1. The container-handling system 100 comprises a holding frame 110 for releasably holding storage containers 50 and a lifting assembly 102 coupled to the holding frame 110 and configured to lower and raise the holding frame 110 relative to the body 27 of the bot 25. Further details of the container-handling system 100 will be described later.

[0057] In FIG. 5 and FIG. 6, a side panel of the bot 25 has been omitted from view to allow the interior of the bot 25 to be seen. The body 27 of the illustrated bot 25 has an upper portion 41 and a lower portion 43. The upper portion 41 is configured to house or support one or more operation components (not shown), such as components of the lifting assembly 102 (e.g. motors), wireless communication components, one or more processors for controlling operation of the bot 25, etc. The lower portion 43 is arranged beneath the upper portion 41. The lower portion 43 is externally open at the bottom and defines a container-receiving space 45 for accommodating at least part of a storage container 50 that has been raised into the container-receiving space 45 by the holding frame 110 and lifting assembly 102. FIG. 5 shows the container-receiving space 45 before it is occupied by a storage container 50 and FIG. 6 shows the container-receiving space 45 after it has been occupied by a storage container 50. The container-receiving space 45 is sized such that enough of a storage container 50 can fit inside the space 45 to enable the bot 25 to move across the track structure 13 on top of storage structure 1 without the underside of the storage container 50 catching on the track structure 13 or another part of the storage structure 1. When the bot 25 has reached its intended destination, the lifting assembly 102 lowers the holding frame 110 and the corresponding storage container 50 out of the container-receiving space 45 and into the intended position. The intended position may be a stack 11 of storage containers 50 or an egress point of the storage structure 1 (or an ingress point of the storage structure 1 if the bot 25 has moved to collect a storage container 50 for storage in the storage structure 1). Although in the illustrated example the upper and lower portions 41, 43 are separated by a physical divider, in other examples, the upper and lower portions 41, 43 may not be physically divided by a specific component or part of the body 27

of the bot **25**. The upper and lower configuration of the bot **25** allows the bot **25** to occupy only a single grid cell **14** on the track structure **13** of the storage system **1**.

[0058] In an alternative example, the container-receiving space **45** of the bot **25** may not be within the body **27** of the bot **25**. For example, the container-receiving space **45** may instead be adjacent to the body **27** of the bot **25**, e.g. in a cantilever arrangement with the weight of the body **27** of the bot **25** counterbalancing the weight of the container **50** to be lifted. In such embodiments, a supporting structure of the lifting assembly **102** may protrude horizontally from the body **27** of the bot **25**, and lifting assembly **102** may be configured to lower and raise the holding frame **110** relative to the supporting structure to lower and raise a storage container **50** into the container-receiving space **45** adjacent to the body **27**.

[0059] FIG. 7 shows an example container-handling system **100** for use with the bot **25**. The container-handling system **100** comprises a holding frame **110** and a lifting assembly **102**.

[0060] The lifting assembly comprises four tethers **104** which are connected at their upper ends to respective spools **106** and at their lower ends to the holding frame **110** (though FIG. 7 shows the tethers **104** disconnected from the holding frame **110**). The tethers **104** may be in the form of cables, ropes, tapes, or any other form of tether with the necessary physical properties to lift the storage containers **50**. The tethers **104** can be wound up or down to raise or lower the holding frame **110** as required. One or more motors **108** or other winding and unwinding means is provided to effect or control rotation of the spools **106** to wind the tethers **104** up or down. The container-handling system **100** is not limited to the particular lifting assembly shown in FIG. 7 and other arrangements suitable for raising and lowering the holding frame **110** may be used.

[0061] The holding frame **110** is in the form of an open rectangular frame. In particular, the holding frame comprises four elongate frame members **112** arranged to form a rectangular frame defining a central rectangular frame opening **114**. In use, the holding frame **110** is orientated such that the frame members **112** lie in a substantially horizontal plane with the frame opening **114** extending vertically through the holding frame **110**.

[0062] The holding frame further comprises a holding assembly **120** configured to engage with engagement features of the storage containers **50** to releasably hold the containers **50** from above. In this example, the holding assembly **120** comprises two gripper assemblies **122** mounted on two frame members **112** at opposing ends of the holding frame **120**.

[0063] FIGS. 8A-8B show a front view of a gripper assembly **122** in isolation. The gripper assembly **122** comprises two grippers **124**. Each gripper **124** comprises a pair of legs **126**. Each leg **126** is pivotally mounted at one end on the frame member **112** for rotational movement in a vertical plane, and each leg **126** comprises a foot **128** at the other end. The legs **126** extend below the frame member **112** such that the feet **128** are located below the frame member **112**. Each pair of legs **126** is rotatable between a closed position (FIG. 8A) in which the feet **128** are close together and an open position in which the feet **128** further apart (FIG. 8B).

[0064] FIGS. 9A-9B show the gripping assembly **122** from the rear when the grippers **124** are in the closed position and the open position respectively. The pairs of legs **126** are driven between the closed position and the open position by a linear actuator **130** and a linkage assembly **132** coupled between the linear actuator and each pair of legs **126**.

[0065] FIG. 10 shows an example storage container **50** that can be held by the holding frame **110**. The storage container **50** comprises a base **51** and sidewalls **52** extending upwards from the base **51** to form a cuboidal storage container with a top opening **56** through which objects can be placed into or removed from the storage container **50**. The storage container **50** further comprises a rim **53** at the top of the sidewalls **52**. The rim **53** comprises engagement features **54** for engaging with the grippers **124** of the holding frame **120**. In this example, the engagement features **54** are apertures extending vertically through the rim **53**.

[0066] FIGS. 11A-11B are side views of the holding frame **110** and the storage container **50** showing how the grippers **124** interact with the apertures **54** to hold and release the storage

container **50**. Each aperture **54** is large enough in the horizontal direction to receive a respective gripper **124** when the gripper **124** is in the closed position but not when the gripper **124** is in the open position. FIG. **11A** shows the holding frame **110** at a position above the storage container **50** in which the grippers **124** are in the closed position and have been received through the apertures **54** until the feet **128** are located below the underside of the rim **53**. FIG. **11B** shows the grippers **124** after they have been moved to the open position. In the open position, the feet **128** of the grippers **124** can engage with the underside of the rim **53** of the storage container **50**. Subsequent raising of the holding frame **110** upwards by the lifting assembly **102** will therefore cause the storage container **50** to be lifted with the holding frame **110**.

[0067] To hold and lift a storage container **50** using the container-handling system **100**, the following operations can take place: the holding frame **110** is lowered by the lifting assembly **102** towards the top of the storage container **50** with the grippers **124** in the closed position until the grippers **124** are received in respective apertures **54** in the rim **53** and the feet of the grippers **124** are below the underside of the rim **53**. The grippers **124** are then moved to the open position such that the feet **128** can engage the underside of the rim **53**. The lifting assembly **102** then raises the holding frame **110**, which causes the feet **28** to engage with the rim **53**, which causes the storage container **50** to be raised with the holding frame **110**. To lower and release a storage container **50**, the above operation is carried out in reverse.

[0068] To facilitate alignment of the holding frame **110** with the storage container **50** so that the grippers **124** are received in the apertures **54** of the storage container **50**, four locating members **116** are mounted at the four corners of the holding frame **110**. The locating members **116** extend downwardly below the frame members **112**. The storage container **50** further comprises cutouts **55** (or other forms of recess) at the corners of the storage container **50** for receiving the locating members **116**. The locating members **116** are configured (e.g. tapered) such that the holding frame **110** can self-align with the storage container **50** when the holding frame **110** is being lowered towards the storage container **50**.

[0069] To allow the container-handling system **100** to determine when a storage container **50** is close enough to the storage container **50** to open the grippers **124**, the holding frame **110** may comprise one or more proximity sensors **117** mounted on the holding frame **110** configured to detect the vertical distance between the holding frame **110** and the top (i.e. the rim **53**) of the storage container **50** as the holding frame **110** is being lowered by the lifting assembly **102**. When the proximity sensor **117** detects that the top of the storage container **50** is at a particular predetermined distance from the holding frame, the lifting assembly **102** can stop lowering the holding frame **110** towards the storage container **50** and the holding assembly **120** can be activated to engage the storage container **50**. Any suitable proximity sensor **117** known in the art may be used, e.g. an ultrasonic sensor.

[0070] FIG. **12** is a perspective view showing holding frame **110** engaged with the storage container **50**. The storage container **50** is occupying a holding region below the holding frame **110**. The holding region is the region which the storage container **50** occupies when the holding assembly **120** is holding the storage container **50**. Put another way, the holding assembly **120** is able to engage with the storage container **50** when the storage container **50** is in the holding region. The holding frame **110** (i.e. the arrangement of the holding members **112**) in this example is sized to have approximately the same horizontal footprint as the storage container **50** such that upwards movement of the storage container **50** with respect to the holding frame **110** is physically limited by the frame members **112**. In particular, the top of the storage container **50** can be located no higher than the bottom of the frame members **112** when being held by the holding frame **110**. Thus, in this example, an approximate upper limit of the holding region may be considered to be a plane coincident with the bottom-most surface of the frame members **112**.

[0071] When the storage container **50** is in the holding region, the frame opening **114** of the holding frame **110** is located above, and is in communication with, the top opening **56** of the

storage container **50** such that objects could be placed into or removed from the storage container **50** via the top opening **56**.

[0072] As mentioned above, the storage container **50** may be used to store objects. A situation may occur in which an overheight object protrudes above the top of the storage container **50**. By providing the holding frame **110** with a frame opening **114**, any overheight objects are likely to protrude through the frame opening **114** when the holding frame **110** is lowered towards the storage container **50**, rather than hitting the holding frame **110**, which could, for example, cause the holding frame **110** to be knocked out of alignment with the storage container **50**.

[0073] The holding frame **110** further comprises an overheight detection system **140** with object detection means for detecting when an object is present in an overheight region **142** (represented in later FIG. **14**). The overheight region **142** is at a fixed location with respect to the holding frame **110** and is above the holding region, i.e. the overheight region **142** is above the top of the storage container **50** when the storage container **50** is being held by the holding assembly **120**. Thus, the overheight detection system **140** can be used to detect when an object is protruding above the top of a storage container **50**. The vertical distance between the overheight region **142** and the holding region may be chosen depending on how high an object can protrude above the top of the storage container **50** before it is considered a problem. The overheight region **142** may be just above the holding region to detect any overheight objects, or the overheight region **142** may be vertically spaced from the holding region to detect any overheight objects which protrude past a threshold height above the holding region.

[0074] FIG. **13** is a simplified schematic diagram showing the holding frame **110** holding the storage container **50** in the holding region. In this example, the object detection means comprises a sensor assembly **143** comprising a plurality of infrared transmitters **144** mounted on a frame member **112** on one side of the frame opening **114** and a plurality of infrared receivers **146** mounted on an opposing frame member **112** on an opposing side of the frame opening **114**. The infrared transmitters **144** are configured to emit infrared light across the frame opening **114** towards the infrared receivers **146** and the infrared receivers **146** are configured to detect infrared light emitted from the infrared transmitters **144**.

[0075] The sensor assembly **143** defines an overheight region **142** (represented in later FIG. **14**) extending between the transmitters **144** and receivers **146**. In this example, the transmitters **144** and receivers **146** are arranged in a substantially horizontal plane within the frame opening **114** and therefore the overheight region **142** in this example is a substantially horizontal plane located within the frame opening **114**.

[0076] FIG. **14** shows a schematic cross-sectional view of the holding frame **110** holding the storage container **50** in the holding region. Given that the frame opening **114** is located above storage container **50** and the sensor assembly **143** is arranged within the frame opening **114**, the overheight region **142** is located directly above the top opening **56** of the storage container **50** when the storage container **50** is in the holding region. FIG. **14** also shows an overheight object **58** located within the storage container **50**. The overheight object **58** protrudes above the top of the storage container **50** such that the overheight object **58** extends through the overheight region **142**, between the infrared transmitters **144** and infrared receivers **146**. This arrangement of the infrared transmitters **144** and infrared receivers **146** can therefore be used to detect the presence of an overheight object within the overheight region **142**.

[0077] The overheight detection system **140** further comprises an object detection controller **162** mounted on the holding frame **110** (shown in FIG. **13**) configured to control the object detection means. The infrared transmitters **144** and infrared receivers **146** are communicably coupled to the object detection controller **162** via electrical wires routed along the frame members **112**. The object detection controller **162** is configured to switch each infrared transmitter **144** on and off. If an infrared receiver **146** detects an infrared signal, a signal is sent from the infrared receiver **146** to the object detection controller **162**.

[0078] When the overheight detection system **140** is active, the object detection controller **162** switches on a first transmitter **144** and waits a period of time to receive a signal from each of the receivers **146** indicating that the infrared light from the first transmitter **144** was detected. If no object is present in the overheight region **142**, each infrared receiver **146** should detect the infrared light emitted from the first infrared transmitter **144** and therefore the object detection controller **162** should receive a signal from each receiver **146**. The object detection controller **162** then switches off the first transmitter **144** and switches on a second transmitter **144** and waits a period of time to receive a signal from each of the receivers **146** indicating that the infrared light from the second transmitter **144** was detected. This process then continues until all of the transmitters **144** have been switched on at least once.

[0079] If, while the overheight detection system **140** is active, an overheight object is within the overheight region **142**, the light path between at least one transmitter **144** and at least one receiver **146** is blocked by the overheight object **58** and therefore at least one receiver **146** will not detect the signal from at least one of the transmitters **144**. The object detection controller **162** will therefore not receive a signal from all of the receivers **146** when one of the transmitters **144** is switched on. The object detection controller **162** is configured to interpret this as meaning that an object **58** is present in the overheight region **142**.

[0080] Depending on the number of transmitters **144** and receivers **146** and the space between them, it is possible that some of the receivers **146** will be too far away to detect the infrared light emitted from a particular transmitter **144**, even if there are no objects in the overheight region. In this case, the object detection controller **162** can be configured to wait to receive a signal from a particular number of receivers **146** (i.e. one or more) after a transmitter **144** has been switched on, or a particular subset of the receivers **146** after a particular transmitter **144** has been switched on. For example, if the transmitters **144** and receivers **146** are arranged such that each transmitter **144** is in the line of sight of only three receivers **146**, then the object detection controller **162** can be configured to wait to receive signals from three receivers **146** after switching on each transmitter

[0081] The number of transmitters **144** and receivers **146** and their arrangement on the holding frame **110** (e.g. the spacing between the transmitters **144** and the spacing between the receivers **146**) can be chosen to reduce the number and size of any blind spots in the overheight region **142**.

[0082] The container-handling system **100** further comprises a frame controller **164** mounted on the holding frame **110** (shown in FIG. 13). The frame controller **164** is communicably coupled to the object detection controller **162** in order to activate the sensor assembly **143** and receive feedback from the object detection controller **162** relating to the outcome of the object detection. The frame controller **164** is further configured to control the holding assembly **120**, i.e. control the actuators **130** to open and close the grippers **124**, and receive feedback from the proximity sensor **117**.

[0083] The above-described arrangement of infrared transmitters **144** and receivers **146** is an example of how photoelectric sensors arranged in a through-beam arrangement can be used to detect the presence of an object in the overheight region **142**. The photoelectric sensors may also be in a retroreflective arrangement instead of a through-beam arrangement. FIG. 15 schematically shows an example in which a plurality of transmitters **144** and receivers **146** are mounted a frame member **112** on the one side of the frame opening **114**. The frame member **112** on the opposite side of the frame opening **114** acts as a reflector **148**, or a separate reflector **148** may be mounted on the frame member **112**. In this example, the sensor assembly **143** is operated in a similar way to the above-described through-beam arrangement, except that the signals emitted by the transmitters **144** are reflected by the reflector **148** before being received by the receivers **146**. If an object blocks the signal between at least one of the transmitters **144** and the reflector **148**, or between the reflector **148** and at least one of the receivers **146**, then at least one of the receivers **146** will not detect a signal, which the object detection controller **162** can use to determine that an object is present in the overheight region **142**.

[0084] The sensor assembly **143** is not limited to comprising infrared photoelectric sensors. Other

types of photoelectric sensors may be used, e.g. visible red or laser. The sensor assembly **143** is also not limited to comprising photoelectric sensors. Other types of object detection sensors known in the art may be used, e.g. ultrasonic sensors, which may be arranged in a through-beam or retroreflective arrangement as described above.

[0085] The sensor assembly **143** does not need to be located within the frame opening **114** and therefore the overheight region **142** does not need to be located within the frame opening **114**. Instead, the sensor assembly **143** may be mounted above or below the frame opening **114** such that the overheight region **142** is above or below the frame opening **114**.

[0086] FIG. **16** is a schematic perspective view of another example of the container-handling system **100** in which the object detection means comprises a camera **150** mounted on the holding frame **110** and one or more processors configured to analyze the images captured by the camera **150**. The one or more processors may be part of the object detection controller **162**. When the overheight detection system **140** is active, the camera **150** is configured to capture at least one image that contains the overheight region **142** and the one or more processors are configured to analyze the captured image to determine whether or not an object is present within the overheight region **142**. As shown in FIG. **16**, the camera **150** may be mounted on the holding frame **110** in one corner of the frame opening **114** and point in an approximately horizontal direction towards the opposite corner of the frame opening **114**. The viewpoint of the camera **150** may be fixed and a region of pixels of each image captured by the camera **150** may be designated as the overheight region **142**. The captured images can then be analyzed by the processor using known image analysis techniques to determine whether or not an object is present in the overheight region **142**. For example, a machine learning algorithm (e.g. a convolutional neural network) trained to detect the presence of an object in the overheight region may be used. The machine learning algorithm may have been trained on images taken by the camera **150** which have been classified into images containing an object in the overheight region and images not containing an object in the overheight region.

[0087] The object detection system **140** may further comprise a light source (e.g. one or more LEDs) mounted on the holding frame **110** that provides light when the camera **150** captures an image so that there is greater contrast between the overheight region **142** and an object within the overheight region **142** in the capture image. This may allow the processor to analyze the captured image and detect objects in the overheight region **142** more effectively. For example, a white light source may be used such that, in the captured image, empty areas of the overheight region appear white and any objects within the overheight region appear dark. The processor may then calculate the proportion of dark pixels to white pixels in the overheight region of the captured image, and if the proportion of dark pixels is above a threshold, the processor determines that an object is present in the overheight region.

[0088] FIG. **17** is a schematic bottom perspective view of an alternative holding frame **110** which does not have a frame opening **114** and instead comprises a substantially horizontal sheet **113** mounted between the frame members **112**. In this example, the horizontal sheet **113** covers the top of the holding frame **110** and the frame members **112** extend downwards from the horizontal sheet **113** to define a recess **115** on the bottom side of the holding frame **110** which provides a headroom for overheight objects above the holding region. In this illustrated example, the overheight detection system **140** is configured to detect the presence of an object within an overheight region **142** located within the recess **115** using the sensor assembly **143** shown in FIG. **13**, i.e. transmitters **144** and receivers **146** mounted on opposing lateral sides of the recess **115**; however, other object detection means such as the sensor assembly **143** shown in FIG. **15** and the camera shown in FIG. **16** may also be used.

[0089] The holding frame **110** shown in FIG. **17** but without the transmitters **144** and receivers **146** also represents another example in which the object detection means can comprise a mechanical force sensor. In particular, the force applied against the underside of the horizontal sheet **113** from

an overheight object when the holding frame **110** is being lowered towards a storage container **50** can be used to determine the presence of the overheight object. In this case, the overheight region **142** is defined by the horizontal sheet **113**, i.e. the plane in which the horizontal sheet **113** lies, or just below the plane in which the horizontal sheet **113** lies. For example, the horizontal sheet **113** may be an elastomeric material under tension and an increase in the tension, as measured by a tension sensor, can be used to determine the presence of an object in the overheight region **142**. Alternatively, the horizontal sheet **113** may be part of a load cell which is configured to detect the load applied to the horizontal sheet **113**. An increase in the detected load (e.g. above a particular threshold) can be used to determine the presence of an object in the overheight region **142**. Although FIG. **17** illustrates the horizontal sheet **113** as forming the top surface of the holding frame **110**, the horizontal sheet **113** could be located at other vertical positions with respect to the frame members **112**. If the holding frame **110** has a frame opening **114**, then the horizontal sheet **113** could be mounted within the frame opening **114**.

[0090] Referring back to FIG. **7**, the container-handling system **100** further comprises an electrical cable **118** to provide power to the various electrical and electronic components on the holding frame **110** (e.g. the frame controller **164**, the holding assembly **120** and the overheight detection system **140**). In this illustrated example, one end of the electrical cable **118** is mounted on a spool that is mounted to a portion of the lifting assembly **102** such that the electrical cable **118** is wound and unwound as the lifting assembly **102** winds and unwinds the tethers **104**. The other end of the electrical cable **118** is mounted to the holding frame **110**, although FIG. **7** shows the electrical cable **118** disconnected from the holding frame **110**. The electrical cable **118** electrically couples a power source of the bot **25** (e.g. a battery) to the electrical and electronic components mounted on the holding frame **110**. The electrical cable **118** also allows for data transfer (e.g. between the frame controller **164** and a bot controller **166** in or on the body **27** of the bot **25**). The electrical cable **118** may be any suitable electrical cable for transmitting power and/or data, e.g. a flexible flat cable (FFC).

[0091] The storage and retrieval system may be used to store products that can be ordered by customers. The products are stored in the storage containers **50** within the storage structure **1** and the identity and location of each product within the storage structure **1** is stored in a database. The location of a product includes the grid cell **14** under which the storage container **50** containing the product is located and the vertical position of the storage container **50** underneath the grid cell **14**.

[0092] FIG. **18** is a block diagram showing an example control system of the storage and retrieval system. The positions of the bots **25** on the track structure **13**, and their routes between positions on the track structure **13**, are controlled by a central control system **170** comprising one or more controllers which are in wireless communication with each bot **25** on the track structure **13** using wireless transmitters and receivers and a suitable wireless communication technology such as 4G, 5G, Wi-Fi, etc. The one or more controllers of the central control system **170** receive data relating to which products are entering the storage and retrieval system and which products have been ordered by customers and uses this data to calculate routes for the bots **25** between target grid cells **14** to pick up and drop off particular storage containers **50** at particular locations. The one or more controllers of the central control system **170** command the bots **25** to move to target grid cells **14** via the calculated routes and pick up or drop off storage containers **50** at the target grid cells **14**.

[0093] The bot **25** comprises a bot controller **166** configured to control functions of the bot **25**, such as operation of the driving assembly to move the bot **25** in a particular direction and for a particular distance, and operation of the lifting assembly **102** to lower and raise the holding frame **110**. As mentioned above, the bot controller **166** is in wireless communication with the central control system **170** via a wireless transmitter and receiver mounted on the bot **25**. The bot controller **166** is configured to receive commands from the central control system **170** (e.g. move to a particular grid cell **14** via a particular route and pick up or drop off a storage container **50**) and send status updates to the central control system **170** (e.g. confirming the position of the bot **25** on

the track structure **13** and the status of the lifting assembly **102**).

[0094] As mentioned above, the holding frame **110** comprises a frame controller **164** configured to control functions of the holding frame **110** including operation of the holding assembly **120**. The bot controller **166** and the frame controller **164** are communicably coupled via the electrical cable **118** such that the bot controller **166** and the frame controller **164** can send signals between each other. For example, the bot controller **166** may command the frame controller **164** to activate the holding assembly **120** and the frame controller **164** may confirm to the bot controller **166** that the holding assembly **120** has been activated.

[0095] As also mentioned above, the holding frame **110** further comprises an object detection controller **162** communicably coupled to the frame controller **164**. The object detection controller **162** is configured to control the object detection means, determine the presence of an object in the overheight region **142** from the objection detection means and communicate the result to the frame controller **164**.

[0096] During operation of the storage and retrieval system, the central control system **170** may command a bot **25** to follow a calculated route to a target grid cell **14** and pick up a target storage container **50** below the target grid cell **14**. The bot controller **166** receives this command and controls the driving assembly to move the bot **25** to the target grid cell **14**. The bot controller **166** then controls the lifting assembly **102** to lower the holding frame **110** from its home position (in or above the container-receiving space **45**) towards the target storage container **50**. Once the proximity sensor **117** detects that the target storage container **50** is in the holding region, the frame controller **164** sends a signal to the bot controller **166**, which controls the lifting assembly **102** to stop lowering the holding frame **110**. While the holding frame **110** is being lowered, the frame controller **164** can activate the overheight detection system **140** via the object detection controller **162** to determine whether or not an object is present in the overheight region **142**.

[0097] If no object is detected within the overheight region **142** during the descent of the holding frame **110**, then the holding frame **110** reaches the top of the target storage container **50** and the frame controller **164** activates the holding assembly **120** to engage and hold the storage container **50**. The frame controller **164** then communicates to the bot controller **166** that the holding assembly **120** is engaged and the bot controller **166** controls the lifting assembly **102** to raise the holding assembly **120** back to its home position such that the target storage container **50** is received into the container-receiving space **45** of the bot **25**.

[0098] If, on the other hand, object detection controller **162** determines that an object is present in the overheight region **142** during the descent of the holding frame **110**, the frame controller **164** will not activate the holding assembly **120** and will communicate the presence of the overheight object to the bot controller **166**, which controls the lifting assembly **102** to halt descent of the holding frame **110** and raise the holding frame **110** back to its home position. Once the holding frame **110** has returned to the home position, the overheight detection system can be reset so that the frame controller **164** can potentially activate the holding assembly **120** the next time the holding frame **110** is lowered to pick up a storage container **50**.

[0099] Alternatively, the overheight detection system **140** may only be activated once the holding frame **110** has been lowered to a position above the storage container **50** at which the holding assembly **120** can engage with the storage container **50** (i.e. when the storage container **50** is in the holding region). If the object detection means detects an object in the overheight region **142**, then the holding assembly **120** is not activated and the lifting assembly **102** raises the holding frame **110** back to its home position. If the object detection means does not detect an object in the overheight region **142**, then the holding assembly **120** is activated to engage and hold the storage container **50** and the lifting assembly **102** raises the holding frame **110** back to its home position such that the storage container **50** is received into the container-receiving space **45**.

[0100] Thus, the container-handling system **100** comprising the overheight detection system **140** can be used to avoid or minimize the risk of any issues associated with the holding frame **110**

attempting to engage with a storage container **50** containing an overheight object, any issues associated with raising a storage container **50** containing an overheight object into the container-receiving space **45** of the bot **25**, and any issues associated with that storage container **50** entering other areas of the storage and retrieval system.

[0101] In addition, the bot controller **166** may send a signal to the central control system **170** to indicate that an overheight object was detected while the bot **25** was at the target grid cell **14**. In response, the central control system **170** may remove that target grid cell **14** from a list of available grid cells **14** of the storage and retrieval system so that no bots **25** are commanded to pick up a storage container **50** from that target grid cell **14**. The central control system **170** may also trigger a process to deal with the overheight object. For example, a special type of bot for dealing with overheight objects may be commanded to move to the target grid cell **14** and retrieve the storage container **50** containing the overheight object. The special bot may comprise a taller container-receiving space **45** that can accommodate a storage container **50** with an overheight object so that the special bot can pick up the storage container **50** containing the overheight object and drop it off at a location where the overheight object can be removed or repositioned within the storage container **50**.

[0102] The overheight detection system **140** can also be used to determine the protruding height of an overheight object, i.e. the vertical height of the portion of the overheight object that is protruding above the top of a storage container **50**. For example, the vertical distance travelled by the holding frame **110** during an overheight time period could be determined, wherein the overheight time period is the time period between the point at which the object detection means first detects an object in the overheight region **142** and the point at which the holding frame **110** is in a position for engaging the storage container **50** (i.e. the point at which the storage container **50** is in the holding region). The vertical distance travelled by the holding frame **110** during the overheight time period will approximately correspond to the protruding height of the overheight object. The vertical distance travelled by the holding frame **110** during the overheight time period could be determined in a number of different ways. For example, the rotation of a component of the lifting assembly that rotates as the tethers **104** unwind (e.g. a motor, a spool or a shaft) could be measured using a rotary encoder and the measurements from the rotary encoder could be translated into the length of tether **104** that was unwound from the spools **106** during the overheight time period, which corresponds to the vertical distance travelled by the holding frame **110**. Alternatively, if the holding frame **110** is lowered (or is assumed to be lowered) at a constant or predetermined velocity, then the vertical distance travelled by the holding frame **110** during the overheight time period could be determined using a predetermined value for the velocity of the holding frame **110**. The determination of the protruding height of an overheight object could be carried out by one or more of the bot controller **166**, the frame controller **164** and the object detection controller **162**.

[0103] By determining the protruding height of an overheight object, the overheight detection system **140** can determine how to proceed based on the protruding height of an overheight object, rather than just the mere presence of an overheight object. For example, if the protruding height is less than a protruding threshold that is unlikely to cause any negative issues, then the holding assembly **120** and lifting assembly **102** may be activated as usual to hold and lift the storage container. In this case, the presence of an overheight object in the storage container **50** may just be recorded in a log for reference. However, if the protruding height is greater than or equal to a protruding threshold, then the holding frame **110** may be raised back to the container-receiving space **45** without activating the holding assembly **120** so that the storage container **50** is not picked up.

[0104] The invention is not limited to the precise forms described above and various modifications and variations will be apparent to the skilled person without departing from the scope of the invention as defined in the accompanying claims.

[0105] For example, instead of holding frame **110** being formed from four separate elongate frame

members **112** joined together, the holding frame **110** may be formed from fewer separate frame members **112** (e.g. two L-shaped frame members), or a single frame member (e.g. a unibody). [0106] Furthermore, the invention is not limited to the particular holding assembly **120** described above and any form of holding assembly **120** for releasably holding a storage container **50** from above may be used. In the case where the holding assembly **120** comprises grippers, the grippers may be moveable between an engaged position for engaging a storage container **50** and a disengaged position for disengaging a storage container **50** using any suitable arrangement of one or more linear or rotational actuators and linkages.

[0107] Furthermore, instead of providing a proximity sensor **117** on the holding frame **110** to detect when the holding frame **110** has been lowered far enough to engage a target storage container **50**, the lifting assembly **102** can lower the holding frame **110** by a predetermined distance based on the known vertical position of the target storage containers **50** within the storage structure **1**.

[0108] Furthermore, the above description of the bot controller **166**, the frame controller **164** and the object detection controller **162** and their associated functions is just one example of how the control system of the bot **25** and the container-handling system **100** may be arranged. In the above-described example, the functions of the container-handling system **100** (e.g. operating the lifting assembly **102**, operating the holding assembly **120** and operating the object detection means) are distributed between the bot controller **166**, the frame controller **164** and the object detection controller **162**. In other examples, the functions of at least two of the bot controller **166**, the frame controller **164** and the object detection controller **162** could be combined. For example, all the functions of the container-handling system **100** may be carried out by a single controller, e.g. the bot controller **166**, or the functions of the object detection controller **162** could be incorporated into the frame controller **164**. The functions of the container-handling system **100** may also be distributed among the bot controller **166**, the frame controller **164**, and the object detection controller in a different way to that described above.

Claims

1. A container-handling system comprising: a holding frame; a holding assembly mounted on the holding frame, wherein the holding assembly is configured to releasably hold a storage container in a holding region with respect to the holding frame; and an overheight detection system comprising object detection means mounted on the holding frame, wherein the object detection means is configured to detect an object in an overheight region, wherein the overheight region is above the holding region.
2. The container-handling system according to claim 1, wherein the overheight region is a substantially horizontal plane, and wherein the object detection means is configured to detect an object between at least two points lying in a horizontal plane.
3. The container-handling system according to claim 1, wherein the holding assembly is configured to releasably hold a storage container from above.
4. The container-handling system according to claim 1, wherein the holding frame comprises a frame opening extending vertically through the holding frame.
5. The container-handling system according to claim 4, wherein at least a portion of the overheight region is located within or above the frame opening.
6. The container-handling system according to claim 4, wherein at least a portion of the object detection means is mounted within or above the frame opening.
7. The container-handling system according to claim 1, wherein the object detection means comprises a sensor assembly arranged on opposing lateral sides of the overheight region.
8. The container-handling system according to claim 7, wherein; the sensor assembly comprises at least one transmitter and at least one receiver mounted on the holding frame on opposing lateral sides of the overheight region, wherein the at least one transmitter is configured to emit a signal

towards the at least one receiver and the at least one receiver is configured to detect the signal from the at least one transmitter; and the overheight detection system further comprises a controller communicably coupled to the at least one transmitter and the at least one receiver, wherein the controller is configured to determine that an object is in the overheight region when the signal between the at least one transmitter and the at least one receiver is interrupted.

9. The container-handling system according to claim 7, wherein; the sensor assembly comprises at least one transmitter and at least one receiver mounted on the holding frame on the same lateral side of the overheight region, and at least one reflector at an opposing lateral side of the overheight region, wherein the at least one transmitter is configured to emit a signal towards the at least one reflector and the at least one receiver is configured to detect the signal from the at least one transmitter after being reflected by the at least one reflector; and the overheight detection system further comprises a controller communicably coupled to the at least one transmitter and the at least one receiver, wherein the controller is configured to determine that an object is in the overheight region when the signal between the at least one transmitter and the at least one receiver via the at least one reflector is interrupted.

10. The container-handling system according to claim 1, wherein the object detection means comprises: a camera mounted on the holding frame, wherein the camera is configured to capture an image containing the overheight region; and one or more processors configured to analyze the image and detect an object within the overheight region.

11. The container-handling system according to claim 1, further comprising a lifting assembly configured to lower and raise the holding frame.

12. The container-handling system according to claim 1, wherein the object detection means comprises a substantially horizontal sheet defining the overheight region and the object detection means is configured to detect a force being applied to the substantially horizontal sheet to detect an object in the overheight region.

13. The container-handling system according to claim 1, further comprising a control system comprising one or more controllers configured to control operation of the holding assembly based on a detection outcome by the object detection means.

14. The container-handling system according to claim 13, wherein the control system is configured such that the holding assembly is not activated if the object detection means detects an object in the overheight region.

15. The container-handling system according to claim 14, wherein the control system is configured to control a lifting assembly to raise the holding frame without activating the holding assembly if the object detection means detects an object in the overheight region.

16. The container-handling system according to claim 13, wherein the control system is configured to determine a height of an object protruding above a storage container based on a vertical distance descended by the holding frame during a time period between a point at which an object is first detected in the overheight region and a point at which a storage container is in the holding region.

17. The container-handling system according to claim 1, wherein: the container-handling system further comprises a load handling device for lifting and moving storage containers arranged in stacks in a storage structure comprising a track structure above the stacks of storage containers, the load handling device comprising: a body; and a driving assembly configured to move the body on the track structure, and the container-handling system further comprises a lifting assembly configured to lower and raise the holding frame of the container-handling system relative to the body of the load handling device.

18. The load handling device according to claim 17, further comprising a container-receiving space configured to accommodate a storage container held by the holding frame, wherein the lifting assembly is configured to raise and lower the holding frame into and out of the container-receiving space respectively.

19. A storage and retrieval system, comprising: a storage structure, comprising: a plurality of stacks

of storage containers, each storage container having a top opening through which objects can be placed into the storage container; and a track structure arranged above the plurality of stacks of storage containers and configured to allow access to each stack from above, and the storage and retrieval system further comprises at least one load handling device, the at least one load handling device comprising: the storage structure, the storage structure comprising a track structure above the stacks of storage containers; a body; a driving assembly configured to move the body on the track structure; and a container-handling system, comprising: a holding frame; a holding assembly mounted on the holding frame, wherein the holding assembly is configured to releasably hold a storage container in a holding region with respect to the holding frame; and an overheight detection system comprising object detection means mounted on the holding frame, wherein the object detection means is configured to detect an object in an overheight region, wherein the overheight region is above the holding region, and a lifting assembly configured to lower and raise the holding frame relative to the body, wherein the container-handling system is configured to pick up and drop off storage containers from and onto the stacks of storage containers.

20. A method of using a load handling device for lifting and moving storage containers arranged in stacks in a storage structure, wherein: the load handling device comprises: the storage structure, the storage structure comprising a track structure above the stacks of storage containers; a body; a driving assembly configured to move the body on the track structure; and a container-handling system, the container-handling system comprising: a holding frame; a holding assembly mounted on the holding frame, wherein the holding assembly is configured to releasably hold a storage container in a holding region with respect to the holding frame; an overheight detection system comprising object detection means mounted on the holding frame, wherein the object detection means is configured to detect an object in an overheight region, wherein the overheight region is above the holding region; and a lifting assembly configured to lower and raise the holding frame relative to the body, and the method comprises: lowering the holding frame towards a storage container; detecting whether or not an object is present within the overheight region; if an object is detected, raising the holding frame away from the storage container without activating the holding assembly; and if an object is not detected, activating the holding assembly to hold the storage container and then raising the holding frame.
